KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2022
Department of Computer Science and Engineering
CSE 4105
Computer Networks

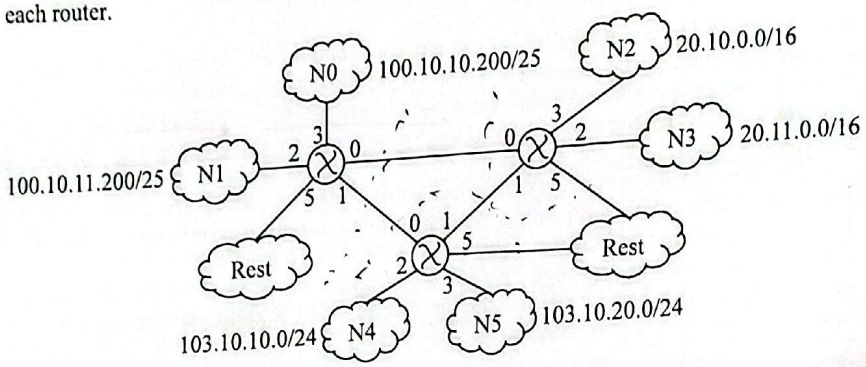
TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define protocol with proper example. Do you think internet standards are necessary for defining computer network protocols? Why or why not? (10)
b) Let you are a network engineer at XYZ Inc. Your task is to design the forwarding tables of each router in a network. Consider the following network and design the forwarding tables of each router. (15)



c) Are there any relations between queuing and switching fabric speed in case of a router? Explain it. (10)

2. a) Define router oscillation problem with proper example. When it occurs? (11)
b) You are given an IPv4 datagram described in hexadecimal format: "4640 001C 00D1 00B9 1E11 DE0B 4AFA 0A14 C878 321E 0000 000F 0272 6F9A". Answer the following questions based on the given datagram (in decimal). (15)
i) Header length (in bytes) = ?
ii) Identifier = ?
iii) Fragment offset = ?
iv) Payload = ?
v) Is it a fragmented packet? If yes, does the packet has any specialty?
vi) Which fields will get updated if the packet is sent to the next hop router?
c) How does classless addressing solve the problem with classful addressing? (09)

3. a) Describe the services offered by network layer. Discuss the necessities of IPv6 over Ipv4. (10)
b) Why does collision occur in CSMA when we only send the packets if the channel sensed idle? Also, demonstrate the collision detection mechanism and algorithm to resolve the issue. (16)
c) "In BGP, packets are not routed to a specific destination address, but instead to CIDRized prefixes"—justify the statement. (09)

4. a) State how packet filtering firewall works. (06)
b) Demonstrate how NAT operates. What are the controversial things behind NAT? (09)
c) Explain how TLS functions. Mention at least one method of obtaining the four keys from the master key. (08)
d) Let you want to send an email to your friend securely. By ensuring the three most important properties of secure communication, demonstrate the scenario for safeguarding email. (12)

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SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) State the advantages of Peer-to-Peer architecture over Client-Server architecture. Despite those advantages, why do we find Client-Server architecture applied in most of the real life cases? (10)

   b) Should your ISP have its own cache server? Justify your answer. Explain how a cache server acts as both a client and a server. (10)

   c) Say, there is a HTTP client sending 1 byte of data every 5 minutes. Let, 1 HTTP packet contains 1 byte of data. The HTTP server uses the data to update its database in real time. Help the network administrator to choose between persistent and non-persistent HTTP connections. Also, advise him to use or avoid pipelining. Explain your reasonings. (15)

6. a) Does rdt 3.0 use same procedure in case of lost packet and lost acknowledgement? Explain with proper illustrations. (10)

   b) Is SMTP enough to build a complete Email architecture? Justify your answer. (10)

   c) Suppose, a client wants to start an online service. He has five computers ready for this job. These are connected to the internet with IP addresses 212.212.212.101–105. He picked "Network utopia.com" as the domain name for the online service. Your job is to make the service accessible by anyone on the internet. What will be your steps to accomplish that? Explain briefly. (15)

7. a) What are port addresses used for? Is there any constraint on how to assign port addresses or how many of them can be assigned at a time? If yes, then discuss about them. (10)

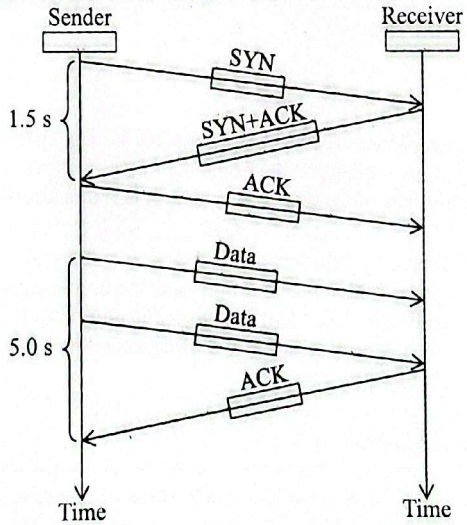
   b) Suppose a sender and a receiver are using a pipelined protocol, i.e., GBN or SR. In this specific case, the rate at which the sender process pushes data is less than the rate at which the receiver process consumes data. We want to introduce a buffer in the receiving side. Is it a wise decision? Explain the reasoning behind your answer. Also, say, the speed of transport layer at sender side is greater than the sender process pushing data to transport layer. In that case, is there a need for a sender buffer at transport layer? Justify your answer. (13)

   c) Let, a proxy server is used for web caching in case of accessing the URL *http://kuet.ac.bd/cse/home.html*, and the browser of client is configured accordingly. Now, explain the scenario of accessing the web page. (12)

8. a) Differentiate between flow control and congestion control. Explain the necessity of either of those in a reliable data transfer protocol. (10)

   b) In the following figure, some desired packet transfer is shown. Introduce a retransmission timer and then calculate and show Retransmission Timeout time (RTO) where applicable. Let, the initial RTO = 6 s. (12)

   Recall that, $RTO = RTT_s + 4RTT_D$, where $RTT_D = (1-\beta)\, RTT_D + \beta \,|\, RTT_s - RTT_M|$. Let, the value of α to calculate RTT_s is 0.125 and β is 0.25.



   c) Suppose, in a TCP connection, the receiver process is slow at data consumption. Eventually, the receiver buffer is full and the receiver advertises zero window size. The sender halts sending. Whenever the receiver process consumes a packet, receiver sends an acknowledgement to the sender. Sender sends the next packet. Will this result in a problem? If yes, discuss about the reasoning and a possible solution. (13)

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